

Bayesian Estimation with Dynare

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Overview

- Can we use information in model solution to estimate parameters?
- Does the data tell us about more than just the mode?
- Can we compute $P(\theta|X)$, the distribution of parameters θ given data?

Bayes Law for Posterior

- Given $P(\theta)$ and data X

$$P(\theta|X) = \frac{P(X|\theta) \cdot P(\theta)}{\int P(X|\theta')P(\theta')d\theta'}$$

- $P(\theta|X)$ is posterior distribution
- $P(\theta)$ is prior distribution, this is a choice
- $P(X|\theta)$ is likelihood function, comes from model solution
- $\int P(X|\theta')P(\theta')d\theta'$ is normalization constant

$$\implies P(\theta|X) \propto L(X|\theta) \cdot P(\theta)$$

Computing the Likelihood

- For a given set of parameters $\hat{\theta}$, dynare 'solve' gives an nth order solution to a model
- If we observe the complete set of variables X in the data, then we can evaluate $L(\hat{X}|\hat{\theta})$ easily
- Problem: we almost never observe the complete set of variables in the data
- How do we compute $L(\hat{X}|\hat{\theta})$ when we observe a restricted set of variables? Filter the data

Computing the Likelihood I

Example

- First order solution to model

$$X_t = D(\theta) \cdot X_{t-1} + E(\theta) \cdot V_t$$

- Sp. we only observe subset of X , y . We can formulate a hidden state representation

$$\text{(state equation)} \quad x_t = A(\theta)x_t + C(\theta)w_{t+1}$$

$$\text{(observation equation)} \quad y_t = G(\theta)x_t + F(\theta)v_t$$

Computing the Likelihood II

- The Kalman filtering equations give

$$\mathbb{E}[y_t | y_{t-1}, y_{t-2}, \dots, x_0]$$
$$y_t - \mathbb{E}[y_t | y_{t-1}, y_{t-2}, \dots, x_0] \sim \mathcal{N}(0, \Sigma_t^y)$$

- Likelihood can be computed from prediction errors
- NB: higher order solutions require particle filter (need newer version of Dynare)

Computing the Posterior

$$\implies P(\theta|X) \propto L(X|\theta) \cdot P(\theta)$$

- Given a choice of θ we can easily calculate $P(\hat{\theta}|X)$
- We can (usually) find $\max_{\theta} P(\theta|X)$ via numerical methods without much difficulty
 - NB: $P(\theta)$ smoothes the posterior, makes maximization easier
- In general $P(\theta|X)$ is an extremely complex object

Drawing from the Posterior

- We are usually interested in calculating objects like

$$\mathbb{E}[g(\theta)] = \frac{\int g(\theta)P(\theta|X)d\theta}{\int P(\theta|X)d\theta}$$

- ie. mean, standard deviation, condifence intervals
- Problem: $P(\theta|X)$ is too complex to draw from
 - Solution: draw from a simple distribution that looks like $P(\theta|X)$

Markov Chain Monte Carlo (MCMC) Method

Metropolis-Hastings Algorithm

- Use a simple heuristic to generate a Markov chain θ such that the distribution of θ converges to $P(\theta|X)$

Algorithm

1. Choose initial draw θ_0
2. For each iteration t
 - Generate candidate draw θ^* by drawing from $N(\theta_{t-1}, \sigma_{mh})$
 - Calculate acceptance ratio $\alpha = P(\theta^*|X)/P(\theta_{t-1}|X)$
 - If $\alpha \geq 1$ set $\theta_t = \theta^*$
 - If $\alpha < 1$ set $\theta_t = \theta^*$ with probability α (generate new draw if not accepted)

MCMC, MH Algorithm

- Choice of σ_{mh} will determine the overall acceptance ratio
 - Brooks & Gellman (1997): optimum acceptance ratio 0.234
- Should generate multiple sequences
- Sequences should behave as if drawn from $P(\theta|X)$, at a minimum the distribution:
 - Should be the same for different parts of same sequence
 - Should be the same for same parts of different sequences

MCMC, MH Algorithm

- Between variance

$$\hat{B} = \frac{1}{J-1} \cdot \sum_{j=1}^J (\bar{\theta}_{\cdot j} - \bar{\theta}_{\cdot\cdot})^2$$

- Within variance

$$\hat{W} = \frac{1}{J} \cdot \sum_{j=1}^J \frac{1}{I-1} \cdot \sum_{i=1}^I (\theta_{ij} - \bar{\theta}_{\cdot j})^2$$

- One check

$$\lim_{I \rightarrow \infty} \hat{B} \rightarrow 0$$

$$\lim_{I \rightarrow \infty} \hat{W} \rightarrow \text{constant}$$

Model

Consider the standard RBC model

```

model;

gamma*c/(1-l) = w;
(1/c) = beta*(1/c(+1))*(1+r(+1)-delta);
c + i = y + g;
k = (1-delta)*k(-1) + i;
w=(1-alpha)*(k(-1)^alpha)*(exp(z)^(1-alpha))*(l^(-alpha));
r=alpha*(k(-1)^(alpha-1))*((exp(z)*l)^(1-alpha));
y = (k(-1)^alpha)*(exp(z)*l)^(1-alpha);
z = rho*z(-1) + e;

end;

```

Declare Parameters to be Estimated & Set Priors

Solve model as usual, after 'stoch_simul' declare parameters to be estimated

```
estimated_params;  
  
alpha, inv_gamma_pdf, .33, .1;  
gamma, normal_pdf, 1, 1;  
  
end;
```

Format for declaring priors:

```
parameter name, prior shape, prior mean, prior std err  
[, 3rd parameter [, 4th parameter]]
```

Set Priors

see Dynare manual for full option set

- Dynare allows the following prior shapes: `beta_pdf`, `gamma_pdf`, `normal_pdf`, `uniform_pdf`, `inv_gamma_pdf`, `inv_gamma1_pdf`, `inv_gamma2_pdf`
- Can also estimate standard errors of shock processes and correlations between variables:
`stderr, e, prior shape, etc...`
`corr, e, y, prior shape, etc...`
- Can choose the initial draw from prior distribution

Set observable variables

After `estimated_params` block set variables that are observable:

```
varobs y;
```

- Cannot have more observed variables than shocks in your model
- Can estimate more parameters than observed variables, but identification tends to be difficult
- Should try to avoid using data series that are highly correlated

Estimation

Estimation command:

```
estimation(datafile=z_ycildata,mh_nblocks=5,mh_replic=10000,  
mh_jscale=.5);
```

You must have data file in the search path

- Dynare can read .m, .mat, or .xls files
- File must have variable names at the beginning of columns, names must match variable names in `mod` file
- Dynare can filter data before estimation (`'prefilter=1'`)

Estimation

`estimation()` options

- `mh_nblocks=5`: Number of MCMC sequences to generate
- `mh_replic=5000`: Number of observations in each sequence
- `mh_jscale=.5`: Scale of jumping distribution in MH algorithm
 - `jscale` $\uparrow \implies$ acceptance ratio $\downarrow$
- `mh_init_scale`: Variance of initial draw in MH algorithm

Dynare Estimation Output I

RESULTS FROM POSTERIOR MAXIMIZATION

parameters

	prior mean	mode	s.d.	t-stat	prior	pstdev
alpha	0.330	0.3827	0.0055	69.8023	invg	0.1000
gamma	1.000	3.6356	0.1214	29.9414	norm	1.0000

Dynare Estimation Output II

ESTIMATION RESULTS

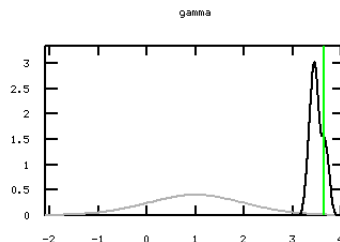
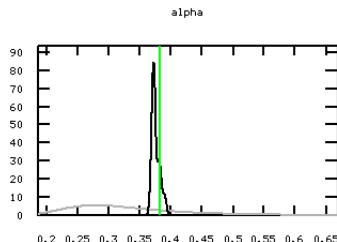
Log data density is -7401.731779.

parameters

	prior mean	post mean	conf. interval	prior pstdev
alpha	0.330	0.3831	0.3740 0.3911	invg 0.1000
gamma	1.000	3.5803	3.4871 3.6886	norm 1.0000



Dynare Estimation Output: Posterior



- Want to see some evidence that your estimator posterior is not being driven by your choice of prior

Dynare Estimation Output: MCMC Diagnostics I

Dynare computes 3 sets of MCMC statistics from Brooks & Gelman (1998)

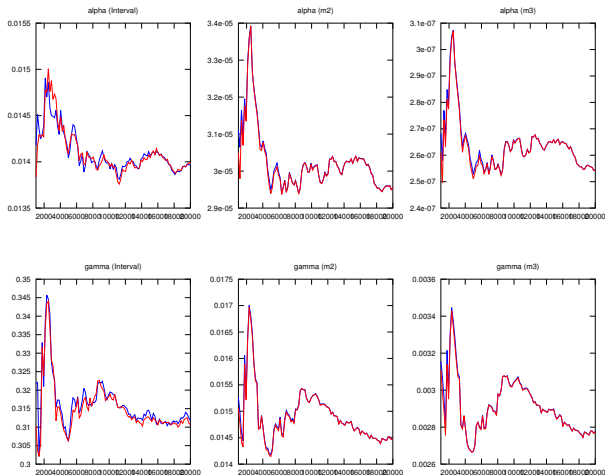
1. Interval: Mean
2. M2: Variance
3. M3: Skewness

For each of these, dynare computes a statistic related to the within-sequence value of each of these (red) and essentially a sum of the within-sequence statistic and a between-sequence variance (blue)

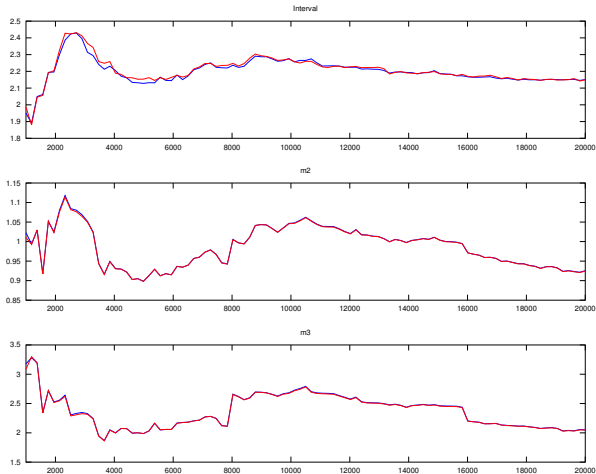
⇒ Want the red and blue to converge and to settle down

Also computes multivariate versions of these statistics

Dynare Estimation Output: MCMC Diagnostics II



Dynare Estimation Output: MCMC Diagnostics III



Bayesian IRF

```
estimation(...,bayesian_irf) varlist
```

- will compute the Bayesian IRFS of all variables in `varlist`
- results stored in structure `oo_.PosteriorIRF.dsge`

Ex: Bayesian IRF, response of c to tfp shock

plot oo_.PosteriorIRF.dsge.Distribution.e

